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SSC-I NATIONAL BOOK FOUNDATION FBISE

MATHEMATICS TEST - CHAPTER 11: BASIC STATISTICS

Name:		Roll No.		Section:

Total Time: 2:30 Hour **Total Marks: 150**
Time Allowed: 150 Minutes

PAPER OBJECTIVE
SECTION-A (MARKS 150)

NOTE: Section-A is compulsory. All parts of this section are to be answered on these pages and handed over to the center superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Q.1 Fill the relevant bubble for each part. All parts carry one mark.

Questions	A	B	C	D	A B C D
1. Find the mode of 2, 2, 11, 2, 5, 7, 12, 6, 7.	2	6	3	1	●○○○
2. The mean of 4 values is 18. 3 values are 26, 2, 6. Find the missing value.	18	42	39	38	○○○●
3. The total angle at the centre of a pie chart is	100°	180°	90°	360°	○○○●
4. Find the mode of the table (x=3, f=2; x=7, f=7; x=9, f=9; x=10, f=8; x=11, f=4).	9	8	11	10	●○○○
5. Find the median of 3, 10, 14, 29, 17.	14.6	14	13	15	○●○○
6. Find the mean of 5, 16, 11, 6, 4, 24.	11	8.5	4	12	●○○○
7. The midpoint of the class 40 – 44 is?	42.5	43.5	42	44	○○●○
8. For n ordered values with n odd, the median is the value at position	(n − 1)/2 + 2	n/2	(n + 1)/2	n + 1	○○●○
9. The number of classes in a table should usually be	always 20	equal to the range	neither too few nor too many	exactly 2	○○●○
10. For grouped data, the mean is given by	∑fx × ∑f	∑x / ∑f	∑f / ∑fx	∑fx / ∑f	○○○●
11. Find the mean of 8, 30, 9, 21, 2.	9	14	2	15	○●○○
12. In a frequency polygon, the polygon is closed by joining ends to	the highest bar	the origin always	the mean	the x-axis at empty classes	○○○●
13. Statistics is the science of	measuring angles	solving equations	collecting, analysing data	drawing shapes only	○○●○
14. The mean of 9 observations is 23. Their sum is?	207	216	32	208	●○○○

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Questions	A	B	C	D	A B C D
15. For classes 42 − 45, 46 − 49, ... the class width is?	4	3	5	8	●○○○
16. Data collected by the investigator himself is called	secondary data	primary data	raw average	grouped class	○●○○
17. The upper class boundary of the class 55 − 64 is?	63.5	65	64	64.5	○○○●
18. The lower class boundary of the class 10 − 19 is?	10	10.5	9	9.5	○○○●
19. For perfectly symmetric data the mean and median are	very different	both zero	equal	both negative	○○●○
20. Frequencies 2, 12, 12, 4 and one missing. If total is 31, the missing is?	31	1	30	2	○●○○
21. In a pie chart, the angle of a sector is	f × 360°	(f / total) × 360°	(total / f) × 360°	(f / total) × 100°	○●○○
22. For classes 32 − 41, 42 − 51, ... the class width is?	10	11	9	20	●○○○
23. The mean of 29, 19, 21, 20, 29, 8 is 21. If each is multiplied by 3, new mean is?	21	64	24	63	○○○●
24. Eye colour of students is an example of	continuous data	qualitative data	quantitative data	discrete data	○●○○
25. The mean of 9 observations is 13. Their sum is?	118	126	22	117	○○○●
26. Smallest value 12, largest 72. Class width for 6 classes ≈ ?	6	60	10	9	○○●○
27. Find the range of 21, 9, 83, 17, 60, 16, 71.	74	75	77	83	●○○○
28. In a pie chart of 30 items, the angle for 6 items is?	73	20	60	72	○○○●
29. The mean of 8, 6, 9, 2, 25 is 10. If 6 is added to each, the new mean is?	10	16	17	60	○●○○
30. Find the mode of 9, 6, 12, 1, 12, 12, 12, 3, 6, 6, 2, 11.	12	8	13	7.5	●○○○
31. A data set having two modes is called	unimodal	trimodal	bimodal	modeless	○○●○

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Questions	A	B	C	D	A B C D
32. A bar chart is mostly used to show	only the mean	only raw data	only continuous data	discrete or categorical data	○○○●
33. Find the median of 41, 36, 13, 44, 25, 31, 39, 6, 27, 4.	29	27	31	30	●○○○
34. Data is grouped into classes when	the data is qualitative	data is large and wide-ranging	all values are equal	only 5 values exist	○●○○
35. Data taken from published records is called	tally data	secondary data	raw mark	primary data	○●○○
36. The upper class boundary of the class 35 − 39 is?	39	40	39.5	38.5	○○●○
37. Find the mode of 4, 3, 3, 8, 6, 1, 12, 2, 5, 3, 2.	4	5	3	2	○○●○
38. The mean of 4 values is 24. A value 44 is added. New mean is?	29	25	28	34	○○●○
39. Frequencies (35−44: 9; 45−54: 6; 55−64: 2; 65−74: 3). The modal class is?	35 − 44	55 − 64	65 − 74	45 − 54	●○○○
40. Find the median of the table (x=2, f=2; x=8, f=8; x=11, f=4; x=14, f=8).	12	8.75	11	8	○○●○
41. In tally marking, the fifth observation is shown by	a horizontal stroke	diagonal stroke across four	a circle	a fifth vertical stroke	○●○○
42. If 5 is added to every value, the mean	increases by 5	stays the same	decreases by 5	is multiplied by 5	●○○○
43. Which average can be found for qualitative data?	median	range	mode	mean	○○●○
44. The mode of a data set is	the average value	the middle value	the range	most frequent value	○○○●
45. Frequencies 15, 14, 8, 3, 11, 10 and one missing. If total is 64, the missing is?	3	64	61	4	●○○○
46. Find the mode of the table (x=1, f=5; x=4, f=11; x=6, f=14; x=10, f=6; x=11, f=4).	10	6	7	11	○●○○
47. The mean of 7 values is 20. 6 values are 19, 23, 3, 25, 8, 15. Find the missing value.	54	20	48	47	○○○●
48. Find the median of 5, 18, 14, 57, 48, 2, 59, 58, 27, 29.	29	28	30	27	○●○○

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Questions	A	B	C	D	A B C D
49. Find the median of the table (x=3, f=3; x=8, f=8; x=9, f=9; x=14, f=8).	8.5	10	8	9	○○○●
50. Find the median of 8, 26, 27, 14, 1, 18, 52.	19	20.86	17	18	○○○●
51. Find the median of 53, 40, 16, 47, 13, 11.	40	29	28	16	○○●○
52. In a pie chart of 180 items, the angle for 2 items is?	5	3.96	4	1.11	○○●○
53. The lower class boundary of the class 45 − 49 is?	45.5	44	45	44.5	○○○●
54. Find the median of 26, 17, 27, 53, 39, 32, 19, 34, 12, 47.	29.5	30.5	32	27	●○○○
55. Frequency of a value is 23 out of 200 observations. Relative frequency?	0.23	8.7	1.12	0.12	○○○●
56. The class boundaries of 20 − 24 are	19 − 25	20 − 24	20.5 − 24.5	19.5 − 24.5	○○○●
57. The midpoint of the class 40 − 49 is?	46	44.5	45	49	○●○○
58. Which is a discrete variable?	Height of a tree	Number of cars in a lot	Time taken to run	Weight of a bag	○●○○
59. Cumulative frequency is	the largest frequency	running total of frequencies	the midpoint	the class width	○●○○
60. The sum of all frequencies in a table equals	total number of observations	the range	the mean	the number of classes	●○○○
61. If every value is multiplied by 3, the mean	stays the same	increases by 3	is divided by 3	is multiplied by 3	○○○●
62. For a frequency polygon, points are plotted against	modes and medians	class limits and mean	boundaries and ranges	class marks and frequencies	○○○●
63. Find the mode of 7, 11, 5, 4, 11, 7, 4, 4, 7, 4, 10.	4	5	7	3	●○○○
64. In a pie chart of 20 items, the angle for 5 items is?	91	90	72	25	○●○○
65. Which is a continuous variable?	Number of books	Number of cars	Number of students	Temperature of a city	○○○●

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Questions	A	B	C	D	A B C D
66. The number of students in a class is an example of a	discrete variable	constant	continuous variable	qualitative variable	●○○○
67. Find the range of 28, 63, 35, 25, 4, 71.	67	66	71	68	●○○○
68. The horizontal axis of a histogram shows	cumulative totals	frequencies	class boundaries	class widths only	○○●○
69. The mean of 10, 9, 17, 17, 6, 12, 15, 34 is 15. If 9 is added to each, the new mean is?	25	15	24	135	○○●○
70. A frequency distribution is a table showing	only the mean	only the range	values with their frequencies	only the mode	○○●○
71. The range of a data set is	the most frequent value	largest − smallest value	largest value + smallest value	the mean of all values	○●○○
72. A frequency polygon is drawn by joining the	lower boundaries	midpoints of tops of bars	bases of the bars	corners of the bars	○●○○
73. Mean of the table (x=6, f=4; x=7, f=3; x=8, f=2; x=10, f=1) is?	7.75	8.1	7.1	17.75	○○●○
74. The median divides ordered data into	two equal halves	ten equal parts	four equal parts	three equal parts	●○○○
75. In the class interval 10 − 19, the lower class limit is	19	9.5	19.5	10	○○○●
76. The mean of 28, 18, 21, 7, 19, 8, 29, 22 is 19. If each is multiplied by 3, new mean is?	57	19	58	22	●○○○
77. Find the median of 9, 15, 31, 36, 42, 40.	34.5	33.5	31	36	○●○○
78. Find the median of 13, 29, 33, 59, 47, 50, 9, 27, 42.	34.33	34	32	33	○○○●
79. The class width of the classes 10 − 19, 20 − 29, 30 − 39 is	20	19	10	9	○○●○
80. The relative frequency of a value is	frequency ÷ total frequency	total ÷ frequency	frequency − total	frequency × total	●○○○
81. Smallest value 6, largest 46. Class width for 5 classes ≈ ?	5	8	40	7	○●○○
82. In a pie chart of 180 items, the angle for 10 items is?	18.95	21	5.56	20	○○○●

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Questions	A	B	C	D	A B C D
83. The weight of a person is a	nominal variable	qualitative variable	discrete variable	continuous variable	○○○●
84. The median is the	sum divided by n	most frequent value	middle value of ordered data	difference of extremes	○○●○
85. Frequencies (30−39: 11; 40−49: 12; 50−59: 6; 60−69: 9). The modal class is?	50 − 59	30 − 39	40 − 49	60 − 69	○○●○
86. In a pie chart of 20 items, the angle for 2 items is?	37	10	32.73	36	○○○●
87. A tally group of five crossed strokes plus two strokes stands for frequency	8	6	5	7	○○○●
88. The mean of 8 values is 19. A value 46 is added. New mean is?	22	32.5	20	23	●○○○
89. Marks obtained in a test are an example of	categorical data	quantitative data	qualitative data	nominal data	○●○○
90. The midpoint of the class 15 − 19 is?	17.5	17	19	18.5	○●○○
91. The most common shoe size sold in a shop is best described by the	range	mode	mean	total	○●○○
92. The midpoint (class mark) of the class 10 − 19 is	15	14	29	14.5	○○○●
93. Find the range of 44, 12, 7, 38, 80, 8, 89, 76, 48.	83	82	89	85	○●○○
94. The mean of 5, 8, 15, 21, 2, 10, 9 is 10. If 2 is added to each, the new mean is?	10	13	12	20	○○●○
95. The lower class boundary of 20 − 24 (width 5) is	20	20.5	19	19.5	○○○●
96. Which average is most affected by extreme values?	none of these	mean	mode	median	○●○○
97. Frequencies 15, 12, 8, 5, 13, 5 and one missing. If total is 62, the missing is?	4	62	5	58	●○○○
98. The number of children in a family is a	discrete variable	continuous variable	qualitative variable	grouped variable	●○○○
99. Find the mean of 21, 2, 28, 18, 4, 5.	13	11.5	14	2	●○○○

Questions	A	B	C	D	A B C D
100. Find the median of 9, 16, 46, 7, 1.	15.8	10	8	9	○○○●
101. Find the median of 50, 40, 22, 54, 19.	41	39	40	37	○○●○
102. Frequencies 3, 7, 3, 6, 13, 3 and one missing. If total is 38, the missing is?	3	38	4	35	●○○○
103. Find the mode of the table (x=1, f=2; x=2, f=10; x=7, f=14; x=10, f=4; x=12, f=5).	10	12	7	8	○○●○
104. Observation means	the mean value	a single value of data	the sum of values	the class width	○●○○
105. Find the range of 85, 20, 54, 61, 27, 6.	80	82	79	85	○○●○
106. Find the mean of 7, 27, 18, 17, 22, 23.	18	19	20	7	○●○○
107. For classes 46 − 55, 56 − 65, ... the class width is?	9	11	10	20	○○●○
108. Find the mean of 17, 2, 23, 4, 16, 22, 14.	16	14	2	15	○●○○
109. The mean of 5 values is 12. 4 values are 29, 13, 3, 10. Find the missing value.	5	12	10	6	●○○○
110. Find the median of 7, 3, 33, 17, 16, 48.	16	17.5	17	16.5	○○○●
111. Frequency of a value is 50 out of 200 observations. Relative frequency?	4	0.25	1.25	0.5	○●○○
112. Mean, median and mode are called measures of	position only	central tendency	frequency	dispersion	○●○○
113. The class boundaries of the class 10 − 19 are	10 − 19	9 − 20	9.5 − 19.5	10.5 − 19.5	○○●○
114. The height of each rectangle in a histogram represents the	class limit	class midpoint	class frequency	class width	○○●○
115. The midpoint of the class 55 − 64 is?	59.5	60	64	61	●○○○
116. The upper class boundary of the class 40 − 49 is?	49.5	50	49	48.5	●○○○

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Questions	A	B	C	D	A B C D
117. Mean of the table (x=1, f=1; x=4, f=4; x=6, f=3; x=10, f=7) is?	7	5.25	26.25	8	●○○○
118. The mean of 30, 7, 12, 21, 12, 30, 14 is 18. If each is multiplied by 3, new mean is?	54	55	18	21	●○○○
119. For an even number of ordered values, the median is	the largest value	the first middle value	the mode	mean of two middle values	○○○●
120. Data collected directly by an investigator and not yet organised is called	raw data	a frequency table	grouped data	secondary data	●○○○
121. The data 3, 5, 7, 9, 11 has	mode 7	no mode	mode 3	two modes	○●○○
122. Frequency of a value is 22 out of 200 observations. Relative frequency?	0.22	0.11	1.11	9.09	○●○○
123. Find the median of 20, 57, 2, 14, 12, 26, 39, 42, 37.	25	26	27.67	27	○●○○
124. The mean of ungrouped data is	sum of values ÷ n	largest − smallest	most frequent value	middle value	●○○○
125. The frequency of a value is	the value itself	the largest value	number of times it occurs	the sum of all values	○○●○
126. For classes 42 − 47, 48 − 53, … the class width is?	5	6	12	7	○●○○
127. Smallest value 13, largest 58. Class width for 6 classes ≈ ?	45	9	8	6	○○●○
128. Find the mean of 19, 28, 4, 24, 10, 12, 26, 13.	17	4	16	18	●○○○
129. In a histogram, the modal class is the class with the	shortest bar	middle bar	widest bar	tallest bar	○○○●
130. Find the mean of 12, 16, 20, 28, 8, 18, 9, 25.	17	18	16	8	●○○○
131. The sum of deviations of all values from their mean is	one	the range	zero	the variance	○○●○
132. Find the median of 4, 3, 12, 19, 24, 34.	12	16.5	19	15.5	○○○●
133. Frequencies (15−24: 12; 25−34: 11; 35−44: 3; 45−54: 6). The modal class is?	15 − 24	25 − 34	45 − 54	35 − 44	●○○○

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Questions	A	B	C	D	A B C D
134. Mean of the table (x=1, f=4; x=2, f=6; x=4, f=6; x=7, f=2) is?	4	13.5	3	3.5	○○●○
135. Class mark is another name for	class boundary	class frequency	class midpoint	class width	○○●○
136. Mean of the table (x=3, f=2; x=4, f=1; x=5, f=3; x=10, f=8) is?	7.5	8.5	26.25	5.5	●○○○
137. Find the mean of 21, 14, 16, 22, 27.	21	14	19	20	○○○●
138. Find the median of the table (x=5, f=6; x=7, f=7; x=9, f=6; x=10, f=9).	7	7.75	9	10	○○●○
139. The mean of 11 observations is 17. Their sum is?	28	187	188	198	○●○○
140. The lower class boundary of the class 25 − 29 is?	24	25.5	24.5	25	○○●○
141. The heights of students in a class form a	categorical variable	discrete variable	qualitative variable	continuous variable	○○○●
142. Find the mean of 29, 19, 29, 26, 16, 16, 18, 31.	24	23	16	22.5	○●○○
143. A histogram consists of	adjacent bars with no gaps	a line joining points only	separated rectangles	a pie divided into sectors	●○○○
144. Find the median of 30, 52, 32, 12, 44.	33	31	34	32	○○○●
145. Frequencies (20−29: 3; 30−39: 10; 40−49: 3; 50−59: 8). The modal class is?	50 − 59	20 − 29	30 − 39	40 − 49	○○●○
146. Frequencies (35−44: 7; 45−54: 6; 55−64: 4; 65−74: 5). The modal class is?	65 − 74	35 − 44	45 − 54	55 − 64	○●○○
147. The mean of 4 values is 22. A value 22 is added. New mean is?	22	23	21	24	●○○○
148. If every value in a data set is 8, then the mean is	0	64	1	8	○○○●
149. A histogram is suitable for displaying	qualitative data only	the median only	continuous grouped data	a single value	○○●○
150. Class width is found by	highest − lowest value	upper − lower boundary	upper limit + lower limit	sum of frequencies	○●○○